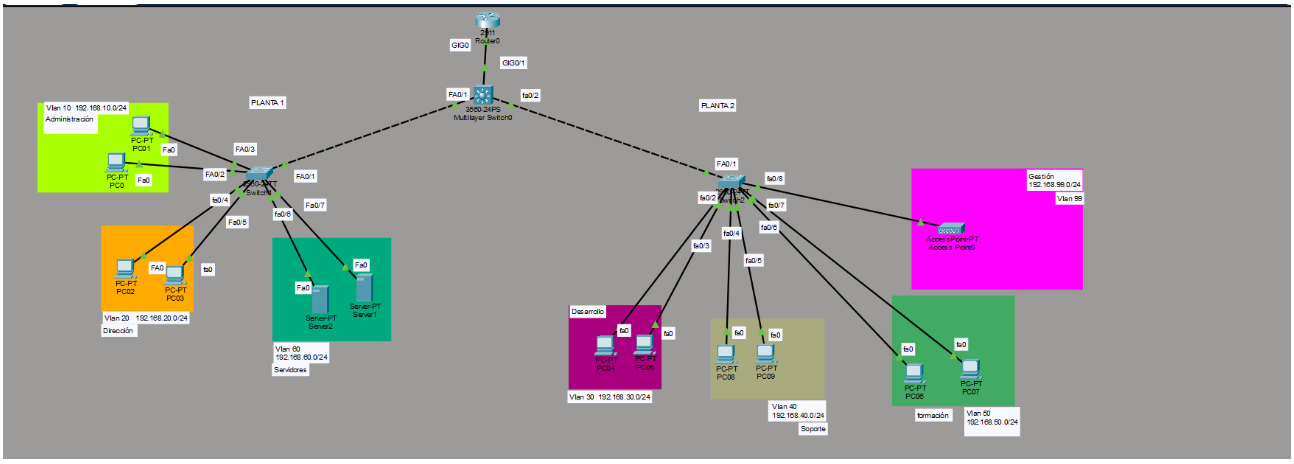
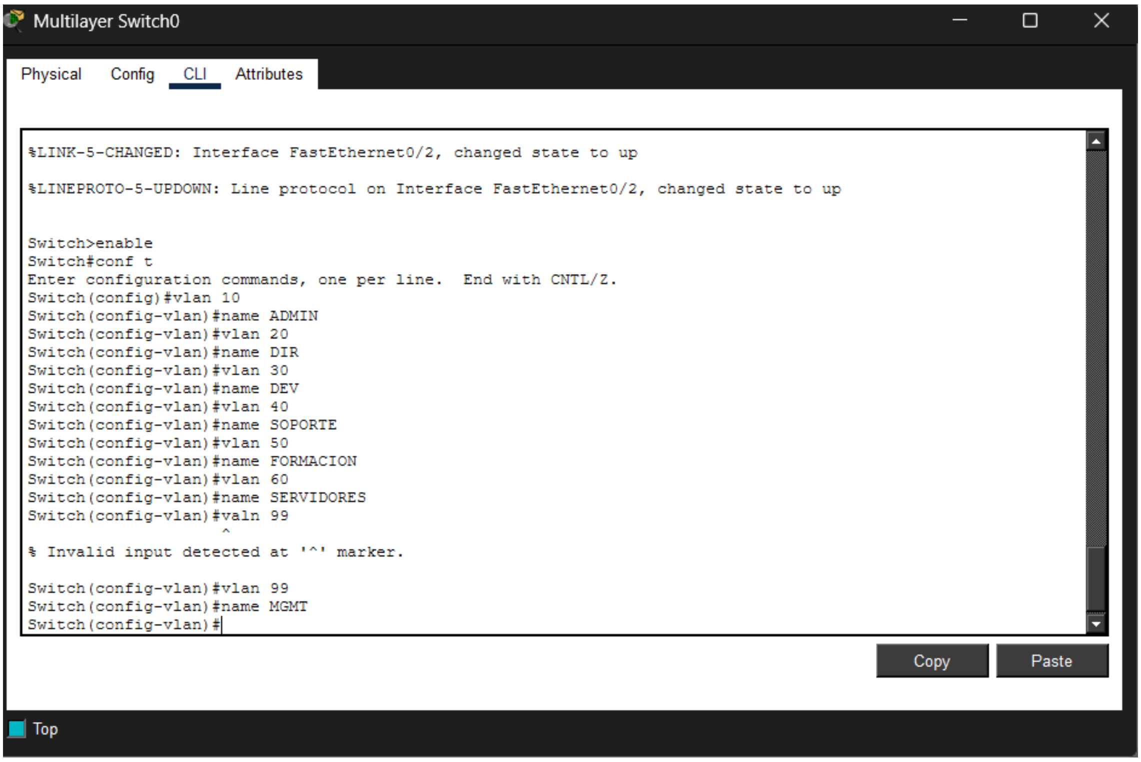
Proyecto intermodular de Administración y planificación de redes.

1. En primer lugar, creo la planificación de la infraestructura de rede en packet tracer.

Decido usar un switch de capa 3 por su facilidad de gestionamiento.



2.Creo las Vlan en el switch de capa 3.



3.Creo las interfaces VLAN en el mismo switch de capa 3

```
Switch#enable
Switch#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#ip routing
Switch(config)#
Switch(config)#interface vlan 10
Switch(config-if)# ip address 192.168.10.1 255.255.255.0
Switch(config-if)# no shutdown
Switch(config-if)#
Switch(config-if)#interface vlan 20
Switch(config-if)# ip address 192.168.20.1 255.255.255.0
Switch(config-if)# no shutdown
Switch(config-if)#
Switch(config-if)#interface vlan 30
Switch(config-if)# ip address 192.168.30.1 255.255.255.0
Switch(config-if)# no shutdown
Switch(config-if)#
Switch(config-if)#interface vlan 40
Switch(config-if)# ip address 192.168.40.1 255.255.255.0
Switch(config-if)# no shutdown
Switch(config-if)#
Switch(config-if)#interface vlan 50
Switch(config-if)# ip address 192.168.50.1 255.255.255.0
Switch(config-if)# no shutdown
Switch(config-if)#
Switch(config-if)#interface vlan 60
Switch(config-if)# ip address 192.168.60.1 255.255.255.0
Switch(config-if)# no shutdown
Switch(config-if)#
Switch(config-if)#interface vlan 99
Switch(config-if)# ip address 192.168.99.1 255.255.255.0
Switch(config-if)# no shutdown
```

4. Configuro los enlaces trunk .

Primero en el switch de capa 3

```
Switch#
Switch#enable
Switch#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#interface fa0/1
Switch(config-if)# switchport mode trunk
Switch(config-if)#
Switch(config-if)#interface fa0/2
Switch(config-if)# switchport mode trunk
```
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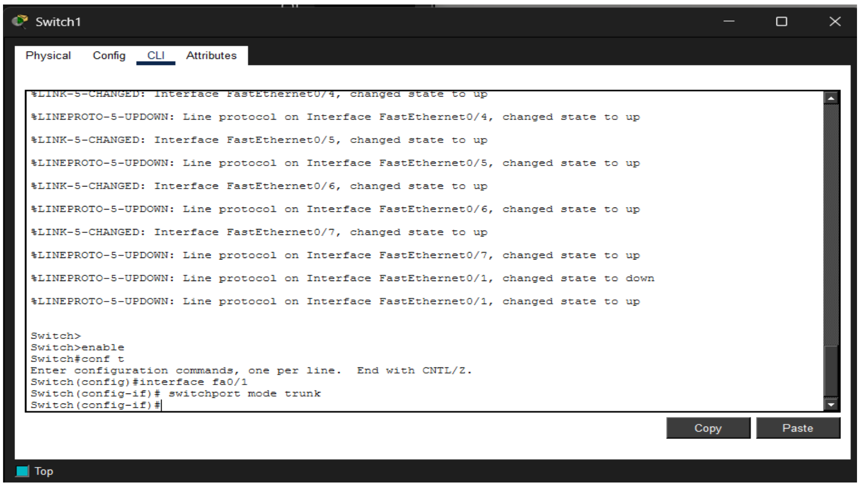
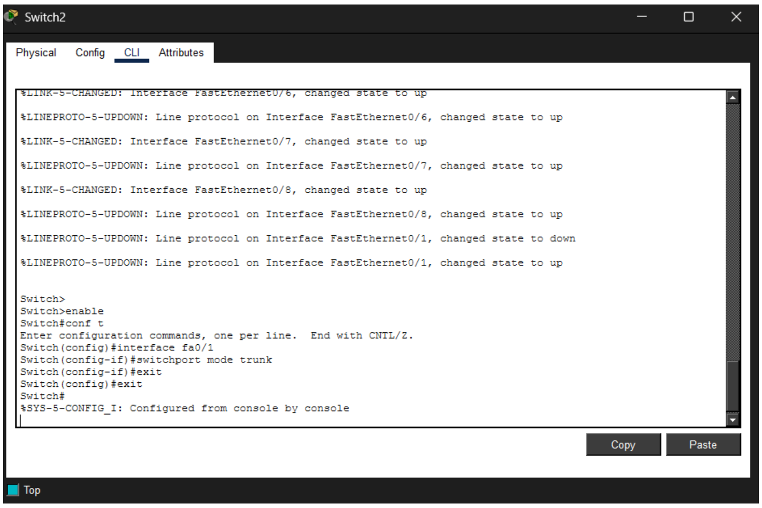
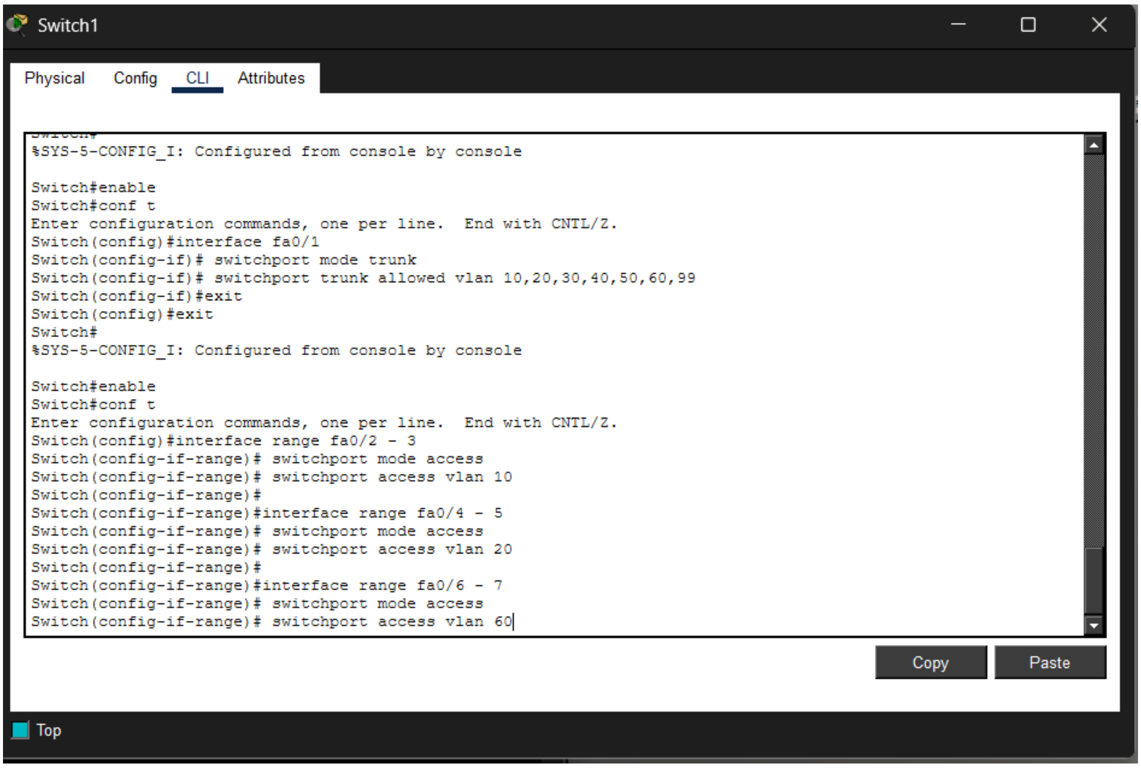
Luego en los switch de plantas.



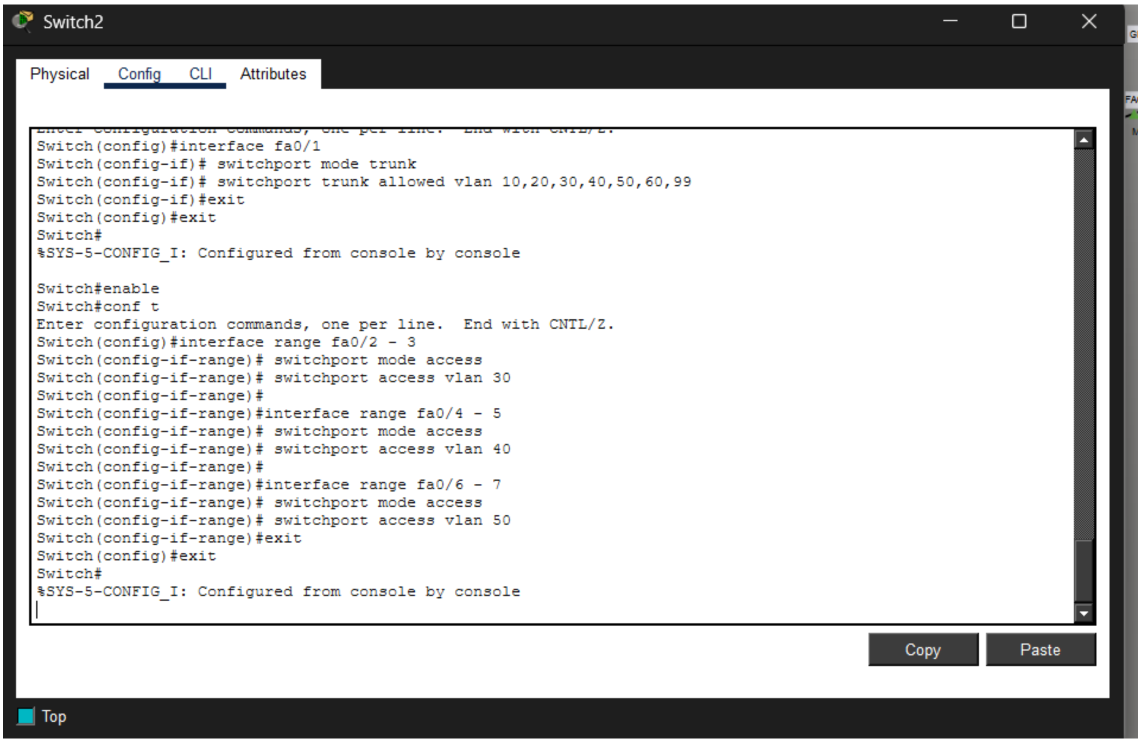


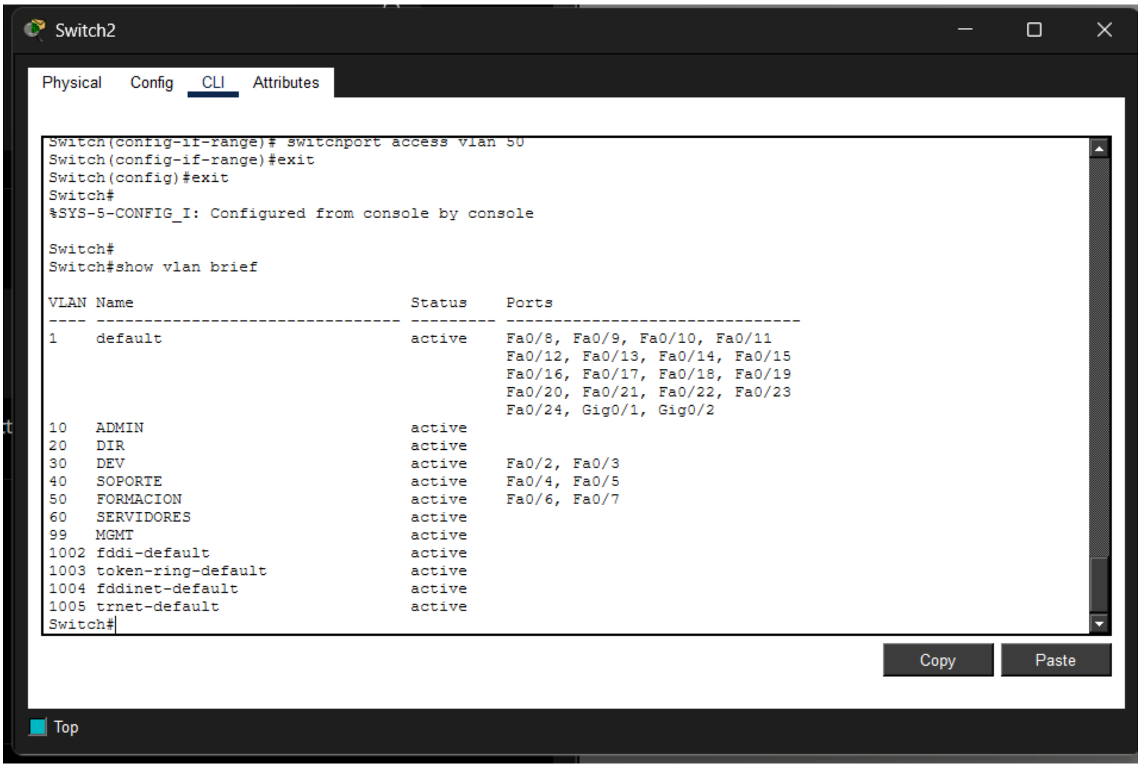
5. Configuro los puertos de acceso.

Primero en las vlan del switch 1



Switch 2





6. Configuro las DHCP en le switch de capa 3

```
Switch>enable
Switch#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#ip dhcp excluded-address 192.168.10.1 192.168.10
Switch(config)#ip dhcp pool ADMIN
Switch(dhcp-config)# network 192.168.10.0 255.255.255.0
Switch(dhcp-config)# default-router 192.168.10.1
Switch(dhcp-config)#
Switch(dhcp-config)#ip dhcp pool DIR
Switch(dhcp-config)# network 192.168.20.0 255.255.255.0
Switch(dhcp-config)# default-router 192.168.20.1
Switch(dhcp-config)#
Switch(dhcp-config)#ip dhcp pool DEV
Switch(dhcp-config)# network 192.168.30.0 255.255.255.0
Switch(dhcp-config)# default-router 192.168.30.1
Switch(dhcp-config)#
Switch(dhcp-config)#ip dhcp pool SOPORTE
Switch(dhcp-config)# network 192.168.40.0 255.255.255.0
Switch(dhcp-config)# default-router 192.168.40.1
Switch(dhcp-config)#
Switch(dhcp-config)#ip dhcp pool FORMACION
Switch(dhcp-config)# network 192.168.50.0 255.255.255.0
Switch(dhcp-config)# default-router 192.168.50.1
Switch(dhcp-config)#
Switch(dhcp-config)#ip dhcp pool SERVIDORES
Switch(dhcp-config)# network 192.168.60.0 255.255.255.0
Switch(dhcp-config)# default-router 192.168.60.1
Switch(dhcp-config)#
Switch(dhcp-config)#ip dhcp pool MGMT
Switch(dhcp-config)# network 192.168.99.0 255.255.255.0
Switch(dhcp-config)# default-router 192.168.99.1
```

7. Creo las Access List. En el switch de capa 3

```
Switch#enable
Switch#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#access-list 100 deny ip 192.168.50.0 0.0.0.255 192.168.20.0 0.0.0.255
Switch(config)#access-list 100 permit ip any any
Switch(config)#
Switch(config)#interface vlan 50
Switch(config-if)# ip access-group 100 in
Switch(config-if)#
Switch(config-if)#access-list 101 deny ip 192.168.30.0 0.0.0.255 192.168.10.0 0.0.0.255
Switch(config)#access-list 101 permit ip any any
Switch(config)#interface vlan 30
Switch(config-if)# ip access-group 101 in
Switch(config-if)#
Switch(config-if)#access-list 102 permit ip 192.168.99.0 0.0.0.255 any
Switch(config)#access-list 102 deny ip any any
Switch(config)#line vty 0 4
Switch(config-line)# access-class 102 in
Switch(config-line)#exit
Switch(config)#exit
Switch#
%SYS-5-CONFIG_I: Configured from console by console
```

Creo las Access list donde formación no puede acceder a dirección.

Desarrollo no puede acceder a admistración

SOLO VLAN MGMT puede administrar.

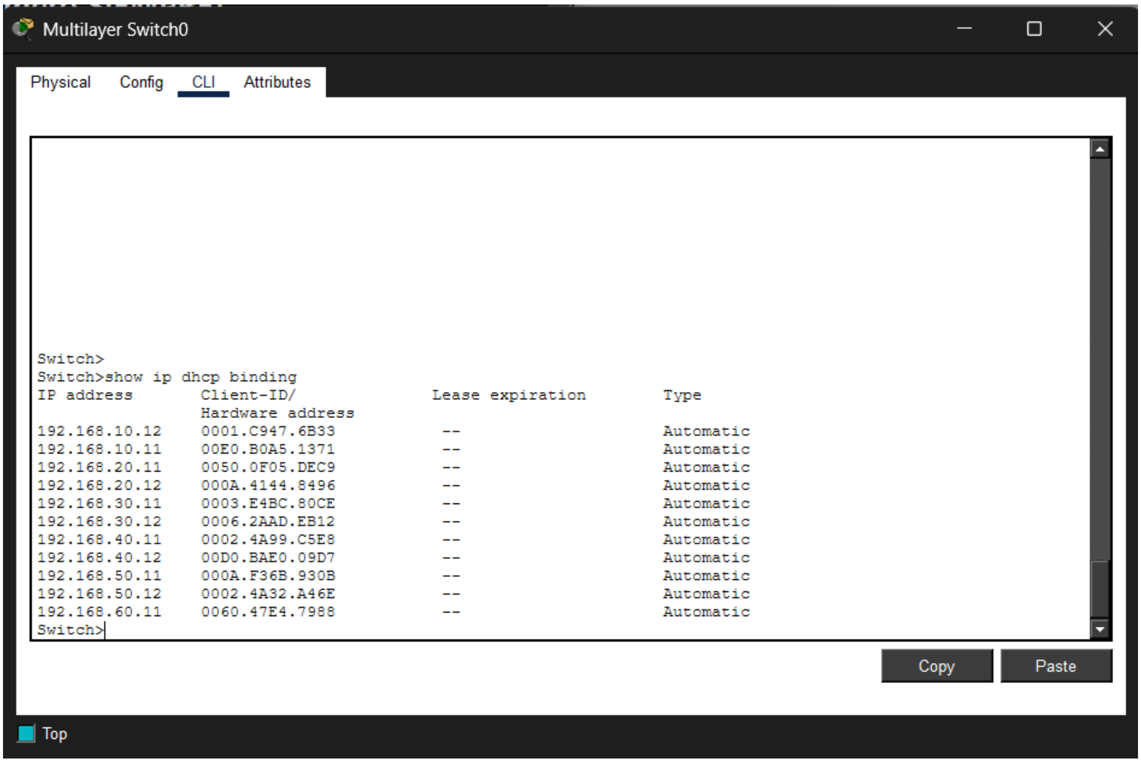
Compruebo.

```
Switch#
Switch#show access-list
Extended IP access list 100
    10 deny ip 192.168.50.0 0.0.0.255 192.168.20.0 0.0.0.255
    20 deny ip 192.168.30.0 0.0.0.255 192.168.10.0 0.0.0.255
    30 permit ip any any
Extended IP access list 101
    10 deny ip 192.168.30.0 0.0.0.255 192.168.10.0 0.0.0.255
    20 permit ip any any
Extended IP access list 102
    10 permit ip 192.168.99.0 0.0.0.255 any
    20 deny ip any any

Switch#show run
Building configuration...

Current configuration : 3481 bytes
!
version 12.2(37)SE1
no service timestamps log datetime msec
no service timestamps debug datetime msec
no service password-encryption
!
hostname Switch
!
!
no profinet
!
!
ip dhcp excluded-address 192.168.10.1 192.168.10.10
ip dhcp excluded-address 192.168.20.1 192.168.20.10
ip dhcp excluded-address 192.168.30.1 192.168.30.10
ip dhcp excluded-address 192.168.40.1 192.168.40.10
ip dhcp excluded-address 192.168.50.1 192.168.50.10
ip dhcp excluded-address 192.168.60.1 192.168.60.10
!
 --More--
```

8. Configuro el wifi en el switch de planta primera.

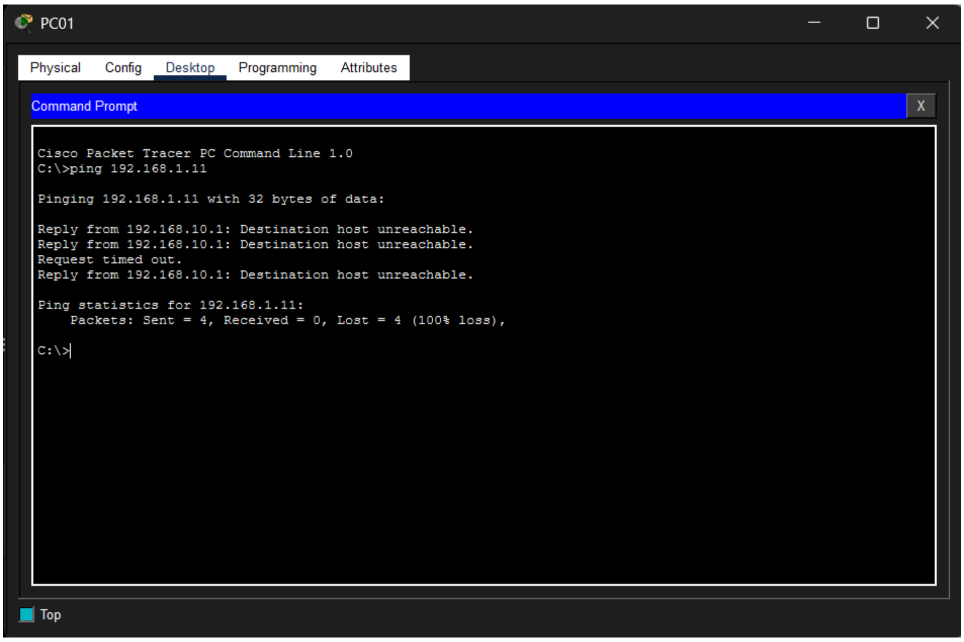
9. Realizo pings para comprobar concectividad, accesos perimitidos y DHCP funcionando.

--Compruebo DHCP en switch l3.

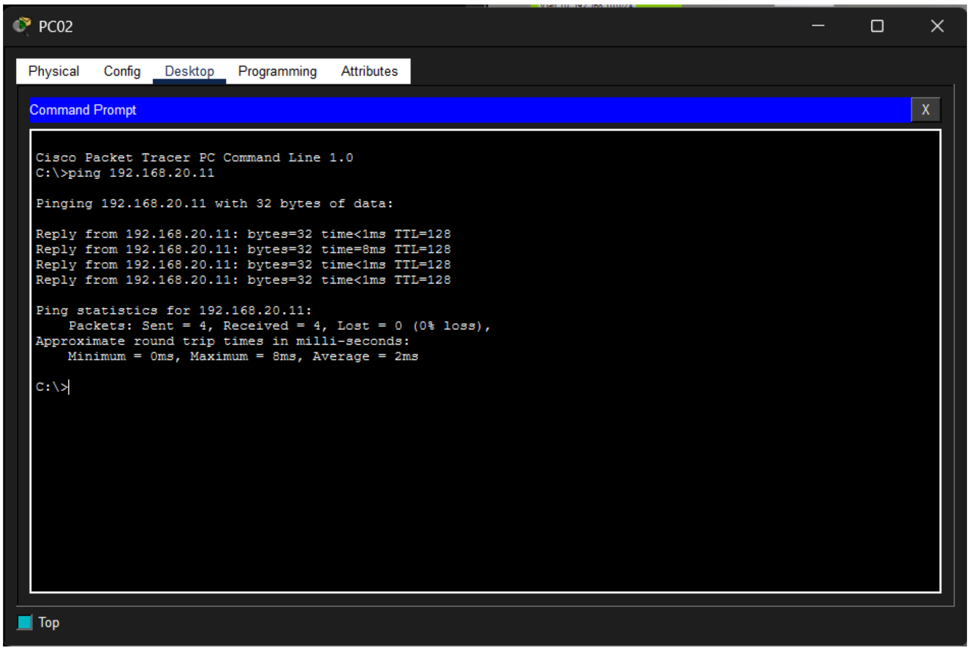


--Hago ping dentro de una misma vlan

(Administracion)



(Dirección)



El resto de ping funcionan perfectamente.

Fin del trabajo